MATH-18A College Algebra **Exam 2 Practice**

The exam will have around 7 questions, and one optional bonus question on the last page. You are allowed a non-graphing calculator and ONE 3×5 notecard. **Show all your work** and **box your answers**.

Section 2.1 Multiply and combine like terms.

1. $f(x) = (3x + 2) \cdot (2x^3 - 2x^2 + 4)$ (4 points)

2. $g(x) = (x - 3) \cdot (x^2 + 3x + 4)$ (4 points)

Section 2.2

3. (6 points) Find **all real solutions** of $15x^2 + 7x - 2 = 0$. Your final answer should include the factored form of the equation AND the x-ints.

For more practice, repeat with the following:

4. $2x^2 - 5x - 3 = 0$

5. $4x^2 - 8x + 53 = 0$

6. $3x^2 - 5x - 3 = 0$

Section 2.3

7. (6 points) Find **all real solutions** to $\dfrac{x}{x+1} - 5 = \dfrac{3}{x}$. State which values of x are not allowed, including any extraneous solutions.

More more practice, repeat with the following:

8. $\dfrac{1}{3} + \dfrac{1}{x-3} = \dfrac{3}{x(x-3)}$

9. $\dfrac{3}{x-3} + \dfrac{x}{x-3} = \dfrac{3}{2}$

10. $\dfrac{3}{x+1} + \dfrac{2}{x-4} = \dfrac{4x-11}{(x+1)(x-4)}$

Section 2.4

11. (6 points) Write the quadratic function $f(x) = 9 + 6x - 3x^2+$ in **vertex form**.

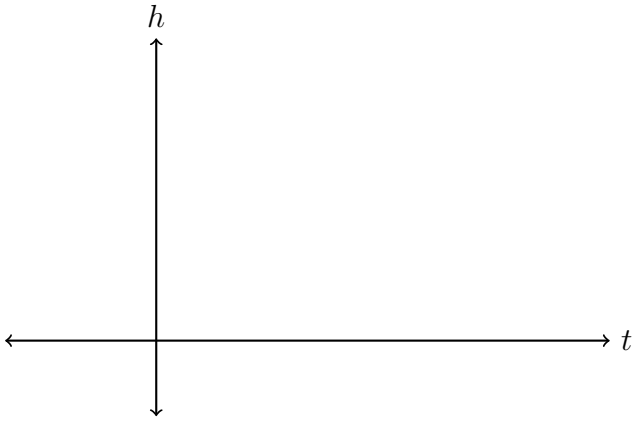
More more practice, repeat with the following:

12. $f(x) = -2x^2 + 12x + 24$

13. $f(x) = 2x^2 + 6x + 1$

Section 2.5

14. (6 points) A ball is launched from atop a 101-foot cliff with an initial velocity of 116 ft/s. Round to the nearest tenth of a second. Make sure to include units in your answers. (Note: if a is not given in the problem, use -16 ft/s^2 or -4.5 m/s^2 depending on the units in the problem.)



a) What is the **height** above the ground when the object is launched?

b) How **long** before the object hits the ground after launch?

c) What is the **maximum height** of the object?

d) **When** does the object reach its maximum height?

More more practice, repeat with the following:

15. A water balloon is catapulted into the air so that its height h, in meters, after t seconds is $h(t) = -5t^2 + 30t + 10$.

16. The height h in feet of a projectile launched vertically upward from the top of a 96-foot-tall tower when time $t = 0$ is given by $h(t) = 96 + 80t - 16t^2$.

Section 2.6

17. (6 points) Divide $f(x) = 3x^3 - 7x^2 - 18x - 8$ by $d(x) = 3x + 2$ using long division. Make sure to **check your answer**, and express your final answer in **division algorithm format**: "dividend = (divisor)(quotient) + remainder".

More more practice, repeat with the following:

18. $f(x) = x^4 + 5x + 2$ by $d(x) = x - 2$

19. $f(x) = 3x^3 - 7x^2 - 18x - 8$ by $d(x) = x + 1$

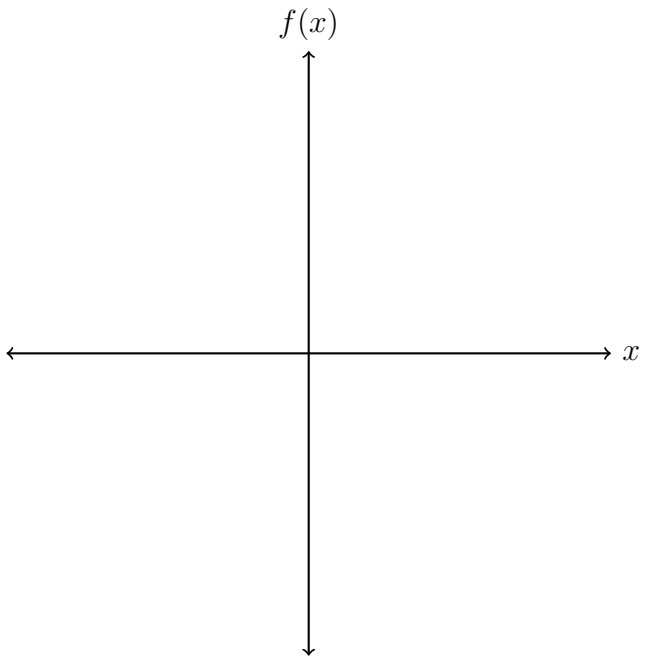
Section 2.7

20. (6 points) Consider the polynomial function $f(x) = (-2x + 12)(x - 1)(x - 7)$. **Sketch** the graph of $f(x)$ on the axes below. You do not need to draw it to scale, but clearly label the zeros of $f(x)$ and clearly show the end behavior of the graph.

a) What is the **end behavior**?

b) What are the **real zeroes**?

c) What is the ***y*-intercept**?



More more practice, repeat with:

21. $f(x) = (x - 3)(x + 1)(x + 4)^3$

22. $f(x) = (x - 2)(x - 1)^2(3x - 7)$